Service	Purpose
Azure Policy	Governance & compliance
Microsoft Defender for Cloud	Security posture + threat protectic
Azure RBAC	Access permissions
Conditional Access	Sign-in control
Key Vault	Secrets, keys, certificates

Keywords ⭐⭐⭐⭐⭐

If you see:

```
Security                    posture
Secure                        Score
Threat                    protection
Security                      alerts
Vulnerability             assessment
JIT              VM           Access
Hybrid                      security
AWS                       protection
GCP protection
```

👉 Answer:

```
Microsoft Defender for Cloud
```

```
Microsoft Defender for Cloud

ASSESS
→ Vulnerability Assessment

SECURE
→ Secure Score
→ Security Recommendations

DEFEND
→ Threat Detection
→ Security Alerts
→ JIT VM Access

Works with:
Azure + On-Prem + AWS + GCP
```

Azure Management and Governance

Cost Management in Azure:

Factors that can affect costs in Azure:

Remember this list:

Resource	Type
Consumption	
Maintenance	
Geography	
Subscription	Type
Azure Marketplace	

If an exam asks:
"What factors influence Azure costs?"
These are the answers.

1. Resource Type ★★★★

Different Azure resources have different prices.
Examples:
Storage
Cost depends on:

- Storage type
- Redundancy option
- Region
- Performance tier

Virtual Machine
Cost depends on:

- VM size
- Number of CPUs
- RAM
- OS licensing
- Storage attached

Larger Resource = Higher Cost

Example:
B-Series VM < D-Series VM < M-Series VM
(cost generally increases)

2. Consumption (Pay for What You Use) ★★★★★

One of the biggest Azure pricing concepts.
Azure uses:
Pay-As-You-Go
You pay only for resources consumed.
Example:

- Use more storage → pay more
- Use more CPU → pay more
- Use less resources → pay less

Azure Pricing Options ★★★★★

A. Pay-As-You-Go

No commitment.
Best for:
Unpredictable workloads
Pros:
☑ Flexible
Cons:
✗ Highest regular pricing

B. Reservations
Commit for:
1 year
or
3 years
Best for:
Predictable workloads
Savings:
Up to 72% discount
Exam Question
Which pricing option offers the largest discount for predictable workloads?
☑ Reservations

C. Azure Savings Plan
Commit to:
Hourly spend amount
instead of a specific VM.
Best for:
Flexible compute workloads
Savings:
Up to 65%

D. Spot VMs ★★★★★
Uses unused Azure capacity.
Best for:
Interruptible workloads
Examples:
- Batch jobs
- Testing
- Data processing

Savings:
Up to 90%
Important
Azure can reclaim the VM anytime.

Exam Question
Which Azure VM option provides the lowest cost but can be interrupted at any time?
☑ Azure Spot VM

3. Maintenance ★★★★

Unused resources still cost money.
Example:
You create:
VM
+
Disk
+
Public IP
+
Storage
Delete only VM:
✗ Disk may still be billed
✗ Public IP may still be billed

Cost Optimization Rule
Delete unused resources

4. Geography ★★★★
Azure pricing varies by region.
Example:
East US ≠ West Europe ≠ Japan East
Different regions have different prices.

Exam Tip
The same VM can cost different amounts depending on:
Azure Region

Network Traffic Costs ★★★★★
Frequently tested.
Inbound Data (Ingress)
Data INTO Azure = Usually FREE

Outbound Data (Egress)
Data OUT OF Azure = Usually CHARGED

Exam Question
Which network traffic is generally free in Azure?
☑ Inbound traffic (Ingress)

Easy
IN = FREE

OUT = PAY

Billing Zones ★★★
Azure uses billing zones for network transfer pricing.
A billing zone is:

Group of Azure Regions
Used For Pricing
Not the same as:
Availability Zones
Exam Question
Are Billing Zones and Availability Zones the same?
❌ No

5. Subscription Type ★★★★
Different subscriptions provide different benefits.
Examples:
Free Account
Includes:

- $200 credit (historically, Microsoft may update amounts)
- Several free services

Enterprise Agreements
May receive different pricing.

Exam Tip
Subscription type can affect:
Cost
Discounts
Credits
Allowances

6. Azure Marketplace ★★★★
Azure Marketplace contains:
Third-Party Solutions
Examples:

- Security software
- Backup tools
- Preconfigured VM images

Important
You pay:
Azure Service Cost
+
Vendor Cost

Exam Question
What additional factor may increase costs when using Azure Marketplace?
☑ Third-party vendor charges

AZ-900 Cost Optimization Questions
Cheapest Option?
For predictable workloads:
☑ Reservations

Largest Savings?
For interruptible workloads:
☑ Spot VM

Reduce Costs?
☑ Remove unused resources
☑ Right-size VMs
☑ Use Reservations

Network Transfer?
☑ Inbound free
☑ Outbound charged

Quick Comparison Table

Option	Commitment	Best For	Savings
Pay-As-You-Go	None	Variable workloads	Standard
Reservation	1 or 3 years	Predictable workloads	Up to 72%
Savings Plan	1 or 3 years	Flexible compute	Up to 65%
Spot VM	None	Interruptible jobs	Up to 90%

Revision ★★★★★

```
Factors Affecting Azure Cost

1. Resource Type
   - VM size
   - Storage type

2. Consumption
   - Pay for usage

3. Maintenance
   - Unused resources still cost money

4. Geography
   - Different regions = different pricing

5. Subscription Type
   - Credits and discounts vary

6. Azure Marketplace
   - Third-party vendor charges
```

```
Pricing Models:
Pay-As-You-Go
Reservations
Savings Plan
Spot VM

Network:
Inbound = Free
Outbound = Charged
```

High-Probability Exam Facts
☑ Spot VM = up to 90% savings, can be interrupted
☑ Reservations = best for predictable workloads
☑ Inbound traffic = free
☑ Outbound traffic = charged
☑ Different Azure regions have different prices
☑ Unused resources can still generate costs

Explore the Pricing Calculator:

Azure Pricing Calculator? ★★★★★
The **Azure Pricing Calculator** is an online tool used to estimate the cost of Azure resources before deploying them.
Purpose
`Estimate Azure Costs Before Deployment`
You can calculate costs for:

- Virtual Machines
- Storage Accounts
- Databases
- Networking
- App Services
- And almost all Azure services

Key Exam Definition ★★★★★
Which Azure tool helps estimate the cost of Azure resources before deployment?
☑ **Azure Pricing Calculator**

Important Note ★★★★
The pricing calculator:
☑ Provides estimates only
☑ Does NOT deploy resources
☑ Does NOT incur charges

Exam Question
Does adding resources to the Azure Pricing Calculator create Azure resources?
✗ No

What Can You Estimate? ★★★★
You can estimate:
Compute

- Virtual Machines
- App Services
- Containers

Storage

- Blob Storage
- Azure Files

Databases

- Azure SQL
- Cosmos DB

Networking

- Bandwidth
- VPN Gateway
- Load Balancer

Factors You Can Configure ★★★★
The calculator allows you to change:

- Azure Region
- VM Size
- Storage Type
- Redundancy Options
- Performance Tier
- Usage Amount

Changing these values changes the estimated cost.

Practical Example ★★★
Suppose you're building a web application:
`Web App`
`+`
`SQL Database`
`+`
`Storage Account`
Add all services to the calculator.
The calculator estimates:
`Monthly Cost` before deployment.

Pricing Calculator vs Actual Billing

★★★★

Pricing Calculator	Actual Azure Bill
Estimate	Real Cost
Before Deployment	After Usage
No Charges	Charges Based on Usage
Planning Tool	Billing Tool

Common AZ-900 Exam Question Scenario:
Your company wants to estimate monthly Azure costs before migrating workloads to Azure.
Which tool should be used?
☑ Azure Pricing Calculator

Frequently Confused Services

Tool	Purpose
Azure Pricing Calculator	Estimate future Azure costs
Azure Cost Management	Monitor and optimize actual spending
Azure Advisor	Recommendations and best practi
Azure Portal	Manage resources

```
Pricing Calculator
=
Before Deployment
=
Estimate Cost

Cost Management
=
After Deployment
=
Track Actual Cost
```

High-Probability Exam Facts

☑ Pricing Calculator estimates Azure costs
☑ No resources are created
☑ No charges occur
☑ Used during planning and budgeting
☑ Can compare costs across regions and service tiers

```
Azure Pricing Calculator

Purpose:
Estimate Azure costs before deployment

Can estimate:
- VMs
- Storage
- Databases
- Networking

Important:
- Estimate only
- No deployment
- No charges

Exam Answer:
"Estimate Azure costs"
→ Azure Pricing Calculator
```

Microsoft Cost Management Tool:

What is Azure Cost Management?

★★★★★

Azure Cost Management helps organizations:
☑ Monitor costs
☑ Analyze spending
☑ Create budgets
☑ Set alerts
☑ Control cloud spending

Simple Exam Definition
Which Azure service helps monitor, analyze, and optimize Azure spending?
☑ **Microsoft Cost Management**

Main Components of Cost Management

★★★★★

Remember:
Cost Analysis
Cost Alerts
Budgets
These are the 3 major features.

1. Cost Analysis ★★★★★

Cost Analysis helps visualize and understand Azure spending.
You can analyze costs by:

- Subscription
- Resource Group
- Resource
- Region
- Service
- Time Period

Example
You discover:
East US VM costs = $500
West Europe VM costs = $200
Cost Analysis helps identify where money is being spent.

Exam Question
Which Cost Management feature helps identify spending trends?
☑ Cost Analysis

2. Cost Alerts ★★★★★
Cost Alerts notify you when spending reaches a threshold.
Types:
Budget Alerts
Notify when spending approaches budget.
Example:
Budget = $1000

Alert at:
80%
90%
100%

Credit Alerts
Used when Azure credits are being consumed.
Example:

- Azure Free Account
- Enterprise Agreement credits

Alert when credit balance is running low.

Department Spending Quota Alerts
Specific to:
Enterprise Agreement (EA)
Alerts when department spending reaches a configured threshold.

3. Budgets ★★★★★
Most important exam topic.
A budget is:
A spending limit
that you define.
Budgets can be created for:

- Subscription
- Resource Group
- Service
- Other scopes

Example
Monthly Budget:
$1000
Alert levels:
80%
90%
100%
When spending reaches those values:
☑ Notifications generated

Important Exam Point
Budgets:
☑ Generate alerts
❌ Do NOT automatically stop spending by default
Many students get trapped here.

Budget + Automation ★★★★
You can connect budgets with automation.
Example:
Budget Threshold Reached
↓
Automation Triggered
↓
Stop Non-Production VM
This requires additional configuration.

Exam Question
Does creating a budget automatically prevent overspending?
❌ No
It only generates alerts unless automation is configured.

Cost Management vs Pricing Calculator ★★★★★
Very common AZ-900 comparison.

Tool	Purpose
Pricing Calculator	Estimate future costs
Cost Management	Monitor actual spending

Memory Trick

```
Before Deployment = Pricing Calculator

After Deployment = Cost Management
```

Cost Management Use Cases ★★★★

Identify Cost Spikes

Example:

```
Yesterday = $50

Today = $500
```

Cost Analysis helps identify the cause.

Validate Budgets

Check whether spending remains within limits.

Find Idle Resources

Discover:

- Unused VMs
- Unused disks
- Overprovisioned resources

Exam Scenarios

Scenario 1

Your company wants to receive an email when cloud spending reaches 80% of the monthly budget.

☑ Budget Alert

Scenario 2

Your company wants to view spending trends across subscriptions and regions.

☑ Cost Analysis

Scenario 3

Your organization wants to monitor actual Azure spending after deployment.

☑ Microsoft Cost Management

Scenario 4

Your company wants to estimate Azure costs before deployment.

☑ Pricing Calculator

Quick Comparison

Feature	Purpose
Cost Analysis	Analyze spending
Budget	Define spending limit
Budget Alert	Notify when budget threshold reached
Credit Alert	Notify when credits are low
Department Spending Alert	EA spending notification

High-Probability Exam Facts

★★★★★

☑ Cost Management = monitor and optimize Azure spending

☑ Cost Analysis = spending trends and visualization

☑ Budget = spending target

☑ Budget Alerts = notifications

☑ Credit Alerts = low Azure credits

☑ Pricing Calculator = estimatesfuture costs

☑ Cost Management = tracks actual costs

```
Microsoft Cost Management

Features:
1. Cost Analysis
   - View spending trends

2. Cost Alerts
   - Budget Alerts
   - Credit Alerts
   - Department Quota Alerts

3. Budgets
   - Set spending limits
   - Generate alerts

Remember:
Pricing Calculator
= Estimate Costs

Cost Management
= Monitor Actual Costs
```

Exam-Favourite One-Liner

Pricing Calculator estimates costs before deployment; Cost Management monitors and controls costs after deployment.

The Purpose of tags:

What Are Azure Tags? ★★★★★
Tags are:
`Name + Value Pairs Attached to Azure Resources`
They provide metadata (extra information) about resources.
Example:

Tag Name	Tag Value
Environment	Production
Owner	John
CostCenter	Finance
AppName	PayrollApp

Simple Exam Definition
What is an Azure tag?
☑ Metadata assigned to Azure resources for organization and management.

Why Use Tags? ★★★★★
Tags help organize resources beyond:
- Resource Groups
- Subscriptions

Think of tags as labels.
Example:
`VM1`
`VM2`
`Storage Account`
`Database`
All can have:
`Environment = Production`
Even if they are in different resource groups.

Major Uses of Tags ★★★★★
Remember:
`Resource Management`
`Cost Management`
`Operations`
`Security`
`Governance`
`Automation`

1. Resource Management ★★★★
Find resources easily.
Example:
`Owner = TeamA`
You can quickly locate all TeamA resources.

Exam Question
Which Azure feature helps organize resources by owner, department, or environment?
☑ Tags

2. Cost Management ★★★★★
Very common exam topic.
Example:
`CostCenter = Finance`
Azure Cost Management can group costs by tag.
This helps:
- Track spending
- Allocate budgets
- Charge departments

Exam Question
Which Azure feature helps allocate costs to different departments?
☑ Tags

3. Operations Management ★★★
Example:
`Impact = MissionCritical`
Helps identify critical resources.
Useful for:
- SLAs
- Operations teams
- Incident response

4. Security Classification ★★★
Example:
`Data Classification = Confidential`
or
`Data Classification = Public`
Helps identify sensitive resources.

5. Governance & Compliance ★★★★
Example:
`Compliance = ISO27001`
Used to identify resources subject to compliance requirements.

6. Automation ★★★
Tags can trigger automation workflows.
Example:
`Environment = Test`

Automation can shut down test resources at night.

Common Tag Examples ★★★★★
Microsoft frequently uses:

Tag Name	Example Value
Environment	Prod / Dev / Test
Owner	John Smith
CostCenter	Finance
AppName	PayrollApp
Impact	High / Medium / Low

Tag Structure ★★★★★
Every tag has:
`Name + Value`
Example:
`Environment = Production`

- Name → Environment
- Value → Production

Important Exam Fact ★★★★★
Tags:
☑ Can be added to resources
☑ Help organize resources
☑ Help track costs
❌ Do NOT inherit automatically from Resource Groups or Subscriptions
This is a favourite exam trap.

Exam Question
If you add a tag to a resource group, do resources inside automatically inherit that tag?
❌ No
Tags are not automatically inherited.

Managing Tags ★★★★
Tags can be:

- Added
- Modified
- Deleted

Using:

- Azure Portal
- Azure CLI
- PowerShell
- ARM Templates
- REST API

Azure Policy and Tags ★★★★
Azure Policy can enforce tagging.
Example:
`Require: Environment Tag`
`Owner Tag`
before deployment.

Exam Question
Which service can enforce that all resources have an Environment tag?
☑ Azure Policy

Tags vs Resource Groups ★★★★★

Resource Group	Tags
Logical container	Metadata labels
Resource belongs to one RG	Resource can have many tags
Used for lifecycle management	Used for organization and reporting

Example
Resource Group:
`Production-RG`
Tags:
`Environment = Prod`
`Owner = FinanceTeam`
`CostCenter = CC100`

Exam Questions
Which Azure feature is used to organize resources by department?
☑ Tags

Question
Which Azure feature helps allocate costs to internal business units?
☑ Tags

Question
What is the format of an Azure tag?
☑ Name-Value Pair

Question
Can Azure Policy enforce tagging rules?
☑ Yes
Question

Do tags automatically inherit from resource groups?

❌ No

```
Azure Tags

Purpose:
Organize Azure Resources

Format:
Name = Value

Examples:
Environment = Prod
Owner = John
CostCenter = Finance

Uses:
✓ Resource Management
✓ Cost Tracking
✓ Security Classification
✓ Governance
✓ Automation

Important:
Tags DO NOT automatically
```

Cost Optimization options in Azure:

What is Cost Optimization? ★★★★★

Cost optimization means:

```
Reduce Azure Costs Without
Sacrificing Required Performance.
```

Azure provides 3 major cost-saving options:

```
Reservations
Azure Savings Plan
Spot Pricing
```

1. Reservations ★★★★★

Reservations are for:

```
Stable
Predictable
Long-running workloads
```

You commit for:

- 1 Year
  OR
- 3 Years

How It Works

Instead of paying normal pay-as-you-go pricing:

```
Commit Capacity
↓
Receive Discount
```

Best Use Case

Examples:

☑ Production VM running 24/7
☑ Production SQL Database
☑ Long-term workloads

Exam Question

Which Azure pricing option is best for predictable workloads?

☑ Reservations

Key Memory

```
Predictable Workload = Reservation
```

2.Azure Savings Plan for Compute

★★★★★

Newer and more flexible than Reservations.

Instead of committing to:

```
Specific VM
```

you commit to:

```
Hourly Compute Spend
```

for:

- 1 year
  OR
- 3 years

Best For

Workloads that:

- Change VM types often
- Scale frequently
- Use multiple compute services

Example

Today:

```
VMs
```

Tomorrow:

```
App Service
```

Next month:

```
Containers
```

Savings Plan still applies.

Exam Question

Which option provides savings while allowing flexibility across Azure compute services?

☑ Azure Savings Plan for Compute

Key Memory
`Flexible Compute = Savings Plan`

3. Spot Pricing (Spot VMs) ★★★★★

Uses:
`Unused Azure Capacity`
at heavily discounted prices.

Advantage
Can save: `Up to 90%`

Disadvantage
Azure can:
`Stop VM`
`Evict VM`
`Reclaim VM`
at any time.

Best For
Interruptible workloads:
☑ Testing
☑ Batch processing
☑ Rendering
☑ Data analytics

NOT Good For
❌ Production databases
❌ Critical applications
❌ Always-on services

Exam Question
Which Azure VM option provides the lowest cost but can be interrupted without notice?
☑ Spot VM

Decision Guide ★★★★★
Choose Reservations When:
`Predictable`
`Long-running`
`Stable`
Examples:

- Production VM
- SQL Database
- ERP System

Choose Savings Plan When:
`Compute usage changes`
`Need flexibility`
Examples:

- Web apps
- API workloads
- Mixed compute services

Choose Spot Pricing When:
`Interruptions are acceptable`
`Lowest cost is priority`
Examples:

- Testing
- Batch jobs
- Rendering

Exam Comparison Table

Feature	Reservations	Savings Plan	Spot VM
Commitment	1 or 3 Years	1 or 3 Years	None
Best For	Predictable workloads	Flexible compute workloads	Interruptible workloads
Cost Savings	High	High	Highest
Azure Can Interrupt?	No	No	Yes
Production Suitable?	Yes	Yes	Usually No

Most Common AZ-900 Questions
Q1A VM runs 24/7 throughout the year. Which pricing option minimizes cost?
☑ Reservation

Q2Compute usage changes frequently across multiple Azure services.
☑ Azure Savings Plan

Q3A workload can tolerate interruptions and requires the lowest possible cost.
☑ Spot VM

Q4Which option allows Azure to reclaim capacity at any time?
☑ Spot VM

```
Reservation
=
Predictable

Savings Plan
=
Flexible

Spot VM
=
Interruptible
```

```
Cost Optimization Options

1. Reservations
   - 1 or 3 years
   - Predictable workloads

2. Azure Savings Plan
   - Commit hourly spend
   - Flexible compute workloads

3. Spot VM
   - Uses unused Azure capacity
   - Up to 90% savings
   - Can be interrupted anytime

Exam Keywords:

Predictable → Reservation

Flexible Compute → Savings Plan

Lowest Cost + Interruptible → Spot VM
```

High-Probability Exam Facts

☑Reservation = stable, predictable workloads

☑ Savings Plan = flexible compute usage

☑ Spot VM = cheapest option

☑ Spot VM can be evicted by Azure

☑ Reservations and Savings Plans require 1-year or 3-year commitments

☑Production workloads typically use Reservations, not Spot VMs.

Features and tools in Azure for governance and compliance:

Purpose of Microsoft Purview

Microsoft Purview?

Microsoft Purview is a **data governance, risk, and compliance solution** that helps organizations:

- Discover data
- Classify sensitive data
- Track data lineage
- Manage compliance
- Govern data across on-premises, Azure, multicloud, and SaaS environments

Simple Definition

Microsoft Purview = Know, Classify, Track, and Protect your data.

Main Components of Microsoft Purview

1. Risk and Compliance

Helps organizations:

- Protect sensitive information
- Manage compliance requirements
- Identify data risks
- Monitor data usage

Examples

- Credit card numbers
- Personal information (PII)
- Financial records
- Healthcare data

Exam Keyword

Protect sensitive data and ensure compliance.

2. Unified Data Governance

Provides visibility into data stored across multiple locations.

Helps organizations:

- Create a data inventory
- Discover sensitive information
- Understand data movement
- Manage access securely

Exam Keyword

Manage and govern data across the entire data estate.

Key Capabilities (Very Important)

Data Discovery

Automatically scans and finds data across:

- Azure
- On-premises
- AWS
- SaaS applications

If the question says **"discover data automatically"**
☑ Microsoft Purview

Data Classification
Identifies and labels sensitive information.
Examples:
- Credit card numbers
- Passport numbers
- Employee records

Exam Tip
If the question says **"classify sensitive data"**
☑ Microsoft Purview

Data Lineage
Tracks where data comes from and where it moves.
Example:
Database → Data Warehouse → Power BI
Purview can show the complete data journey.
Exam Tip
If the question says **"track data movement"** or **"data lineage"**
☑ Microsoft Purview

Benefits of Microsoft Purview
Create a Data Map
Know where organizational data exists.
Find Sensitive Data
Locate confidential information quickly.
Track Data Flow
Understand how data moves between systems.
Secure Data Access
Control access to sensitive information.
Meet Compliance Requirements
Support regulatory standards and audits.

AZ-900 Exam Keywords

Keyword	Service
Data Governance	Microsoft Purview
Data Discovery	Microsoft Purview
Data Classification	Microsoft Purview
Data Lineage	Microsoft Purview
Compliance Management	Microsoft Purview

Exam Questions
Which Azure service helps discover and classify sensitive data?
☑ Microsoft Purview

Which Azure service provides unified data governance?
☑ Microsoft Purview

Which Azure service tracks data lineage?
☑ Microsoft Purview

Which Azure service helps manage compliance and data risks?
☑ Microsoft Purview

Revision
Microsoft Purview
- Discover Data
- Classify Data
- Track Data Lineage
- Manage Compliance
- Govern Data Across Azure, On-Premises, Multicloud & SaaS

Purview = "View & Govern Your Data"
👉 **Discovery + Classification + Lineage + Compliance = Microsoft Purview**

Purpose of Azure Policy:
Azure Policy is a service that helps you **create, assign, and enforce rules** for Azure resources.
It ensures resources remain compliant with organizational standards and governance requirements.
Simple Definition

Azure Policy = Governance and Compliance Service for Azure Resources

Why Use Azure Policy?
Azure Policy helps organizations:
- Enforce standards
- Maintain compliance
- Prevent non-compliant resources
- Audit existing resources
- Automatically remediate issues

Example
You can create a policy that:
- Allows only specific VM sizes
- Requires resource tags
- Requires encryption
- Restricts resource locations

How Azure Policy Works
Step 1: Define Policy
Create a rule.
Example:
- Only allow East US region
- Require encryption on storage accounts

Step 2: Assign Policy
Assign the policy to a:
- Management Group
- Subscription
- Resource Group
- Resource

Step 3: Evaluate Resources
Azure checks whether resources comply with the policy.
Result:
☑ Compliant
❌ Non-Compliant

Step 4: Remediation (Optional)
Azure can automatically fix certain issues.
Example:
Missing tag → Azure automatically adds it.

Policy Scope (Very Important)
Policies can be assigned at:

Scope	Description
Management Group	Multiple subscriptions
Subscription	Entire subscription
Resource Group	Group of resources
Resource	Individual resource

Exam Tip
Policies assigned at higher levels are **inherited** by lower levels.
Example:
Management Group → Subscription → Resource Group → Resource

Azure Policy Initiatives
What is an Initiative?
An Initiative is a collection of related policies grouped together.
Simple Definition
Initiative = Group of Policies

Example
Security Initiative may include:
- Monitor unencrypted SQL databases
- Monitor OS vulnerabilities
- Monitor missing endpoint protection

Instead of assigning each policy separately, assign the initiative once.
Exam Tip
Policy = Single Rule
Initiative = Collection of Policies

Built-In Policies
Azure provides many built-in policies for:
- Compute
- Storage
- Networking
- Security
- Monitoring

Examples:
- Allowed locations
- Allowed VM sizes
- Require tags
- Require encryption

Azure Policy and Compliance

Azure Policy continuously evaluates resources and identifies:
☑ Compliant Resources
❌ Non-Compliant Resources
This helps organizations maintain governance standards.

Azure Policy and AI-Assisted Changes
Even if changes are made using:

- Copilot
- Automation
- Scripts
- Azure DevOps

Azure Policy still enforces organizational rules.
Example
A policy may require:

- Approved VM sizes
- Required tags
- Specific regions
- Security baselines

The deployment will be checked against those policies.

Common Exam Questions
Which Azure service enforces organizational standards and compliance?
☑ Azure Policy

Which Azure service can prevent non-compliant resources from being created?
☑ Azure Policy

Which Azure service continuously evaluates resources for compliance?
☑ Azure Policy

What is a collection of Azure Policies called?
☑ Initiative

Policies assigned at a Management Group are inherited by?
☑ Subscriptions, Resource Groups, and Resources

Azure Policy vs RBAC (Very Important)

Azure Policy	Azure RBAC
Controls what resources can be created/configured	Controls who can access resources
Governance & Compliance	Authorization
Evaluates resource properties	Evaluates user permissions

Exam Favorite
Azure Policy = What can be done
Azure RBAC = Who can do it

Quick Revision
Azure Policy

- Enforce standards
- Ensure compliance
- Audit resources
- Prevent non-compliant deployments
- Auto-remediate issues
- Uses Policies and Initiatives

Policy = Rules for Resources
RBAC = Permissions for Users
👉 **Policy controls resources, RBAC controls access.** 🚀

Purpose of Resource Locks:
Resource Locks help prevent Azure resources from being **accidentally deleted or modified**.
Even if a user has permissions through RBAC, a resource lock can block certain actions.
Simple Definition
Resource Lock = Extra protection against accidental changes or deletion.

Why Use Resource Locks?
Resource locks help protect critical resources such as:

- Virtual Machines
- Storage Accounts
- SQL Databases
- Resource Groups

Example
You have a production database.
Even if an administrator accidentally tries to delete it:
☑ Resource Lock prevents the deletion.

Types of Resource Locks
Azure provides **two types** of locks.

1. Delete Lock (CanNotDelete)
Allows:
☑ Read
☑ Modify
Blocks:
❌ Delete
Example
A Storage Account can still be updated, but cannot be deleted.
Exam Tip
Delete Lock = Prevents deletion only.

2. ReadOnly Lock
Allows:
☑ Read
Blocks:
❌ Modify
❌ Delete
Example
Users can view a VM but cannot make changes or delete it.
Exam Tip
ReadOnly Lock = Most restrictive lock.

Quick Comparison

Action	Delete Lock	ReadOnly Lo
Read	✅	✅
Modify	✅	❌
Delete	❌	❌

Exam Favorite
ReadOnly Lock blocks both modification and deletion.

Lock Scope
Resource Locks can be applied at:
- Subscription
- Resource Group
- Individual Resource

Lock Inheritance (Very Important)
Locks are inherited.
Example
If you place a lock on a:
Resource Group
Then all resources inside inherit the lock:
- VM
- Storage Account
- SQL Database

Exam Tip
Higher-level locks automatically apply to child resources.

Managing Resource Locks
You can create and manage locks using:
- Azure Portal
- Azure PowerShell
- Azure CLI
- ARM Templates

How to Modify a Locked Resource?
Two-Step Process
1. Remove the lock
2. Perform the action

Important
Resource Locks override normal RBAC permissions.
Even an:
☑ Owner
must remove the lock first before performing a blocked operation.

Resource Locks vs RBAC

Resource Lock	RBAC
Protects resources from accidental changes	Controls user permissions
Prevents delete/modify operations	Grants or denies access
Applies to resources	Applies to users/groups

Exam Favourite

RBAC controls WHO can act.
Resource Locks protect WHAT can be changed or deleted.

Common Exam Questions
Which Azure feature prevents accidental deletion of resources?
☑ Resource Locks

Which lock prevents deletion but still allows modifications?
☑ Delete Lock (CanNotDelete)

Which lock prevents both modification and deletion?
☑ ReadOnly Lock

Do Resource Locks override RBAC permissions?
☑ Yes

If a lock is applied to a Resource Group, what happens?
☑ All resources inside inherit the lock.

Revision
Resource Locks
Delete Lock

- Read ☑
- Modify ☑
- Delete ❌

ReadOnly Lock

- Read ☑
- Modify ❌
- Delete ❌

🔒 **Delete Lock = "Can't Delete"**
🔒🔒 **ReadOnly Lock = "Can Only Read"**

Exam Keywords

- Prevent accidental deletion → **Resource Locks**
- CanNotDelete → **Delete Lock**
- Read-only access → **ReadOnly Lock**
- Inherited protection → **Lock Inheritance**
- Overrides RBAC actions → **Resource Locks**

👉 **Resource Locks = Last line of protection for critical Azure resources.**

The Purpose of the Service trust portal:

What is the Service Trust Portal?
The **Microsoft Service Trust Portal (STP)** is a website that provides information about Microsoft's:

- Security practices
- Privacy policies
- Compliance programs
- Audit reports
- Certifications

Simple Definition
Service Trust Portal = Microsoft's Compliance and Security Documentation Center

Why Use the Service Trust Portal?
Organizations use STP to:

- Verify Microsoft compliance certifications
- Review audit reports
- Understand security controls
- Meet regulatory requirements
- Support risk assessments

What Information Does It Provide?
Security Information
Details about how Microsoft protects:

- Azure services
- Microsoft 365 services
- Customer data

Privacy Information
Provides information about:

- Data protection
- Data handling
- Privacy commitments

Compliance Documentation
Access documents related to standards such as:

- ISO 27001
- SOC 1
- SOC 2
- GDPR
- PCI DSS
- FedRAMP

Exam Tip
If the question mentions:
"Compliance reports"
☑ Service Trust Portal

Main Features
My Library
Allows users to:

- Save important compliance documents
- Pin frequently used reports
- Receive update notifications

All Documents
Contains:

- Audit reports
- Certifications
- Compliance guides
- Security documentation

Compliance Manager Resources
Provides resources to help organizations understand and manage compliance requirements.

Access Requirements
To access most documents:

- Sign in with a Microsoft account
- Accept Microsoft's NDA (for some compliance materials)

Common Use Cases
Audits
An auditor wants proof that Azure complies with ISO 27001.
☑ Use Service Trust Portal

Regulatory Reviews
An organization needs Microsoft's compliance reports for internal review.
☑ Use Service Trust Portal

Security Assessments
Security teams need details about Microsoft's security controls.
☑ Use Service Trust Portal

Common Exam Questions
Which Microsoft portal provides compliance reports and audit documentation?
☑ Service Trust Portal

Where can organizations access Microsoft security and compliance documentation?
☑ Service Trust Portal

Which portal provides information about Microsoft's certifications such as ISO, SOC, and GDPR?
☑ Service Trust Portal

Service Trust Portal vs Microsoft Purview

Service	Purpose
Microsoft Purview	Data governance, classification, compliance management
Service Trust Portal	Compliance reports, certifications, audit documentation

Exam Favourite

Purview = Manage Your Data
Service Trust Portal = View Microsoft's Compliance Reports

Revision
Service Trust Portal (STP)

- Security Documentation
- Privacy Information
- Compliance Reports
- Audit Reports
- Certifications
- Regulatory Standards

📋 **Service Trust Portal = Trust Microsoft through Compliance Evidence**
Think:
"Need proof that Microsoft is compliant?" → Service Trust Portal

Exam Keywords

- Audit Reports → **Service Trust Portal**
- Compliance Documentation → **Service Trust Portal**
- Security Reports → **Service Trust Portal**
- ISO/SOC/GDPR Certifications → **Service Trust Portal**
- Microsoft's Compliance Evidence → **Service Trust Portal**

👉 **Service Trust Portal = Microsoft's official repository for security, privacy, and compliance documentation.** 🚀

Features and tools for managing and deploying Azure services:

Why Do We Need Azure Management Tools?

Azure provides multiple tools to create, manage, monitor, and automate cloud resources.

Main Azure Management Tools

- Azure Portal
- Azure Cloud Shell
- Azure PowerShell
- Azure CLI
- Copilot in Azure

Azure Portal

Azure Portal is a **web-based graphical interface (GUI)** for managing Azure resources.

Simple Definition

Azure Portal = Browser-based GUI for Azure

What Can You Do?

- Create resources
- Manage resources
- Monitor resources
- View dashboards
- Configure services

Benefits

- ☑ Easy to use
- ☑ No scripting required
- ☑ Accessible from any browser

Exam Tip

If the question mentions:

"Graphical interface for Azure"

- ☑ Azure Portal

Azure Cloud Shell

Azure Cloud Shell is a **browser-based command-line environment** available directly from Azure Portal.

Simple Definition

Cloud Shell = Command Line in the Browser

Key Features

- No installation required
- Already authenticated with Azure account
- Accessible from Azure Portal
- Supports PowerShell and CLI

Exam Tip

If the question says:

"Browser-based shell with no installation"

- ☑ Azure Cloud Shell

Azure PowerShell

Azure PowerShell is a command-line tool that uses PowerShell cmdlets to manage Azure resources.

Simple Definition

Azure PowerShell = PowerShell-based Azure management tool

Uses

- Create resources
- Manage resources
- Automate tasks
- Deploy infrastructure

Benefits

- ☑ Automation
- ☑ Scripting
- ☑ Repeatable deployments

Example

```
New-AzResourceGroup
```

Creates a Resource Group.

Exam Tip

If the question mentions:

"Cmdlets"

- ☑ Azure PowerShell

Azure CLI

Azure CLI is a cross-platform command-line tool used to manage Azure resources.

Simple Definition

Azure CLI = Bash-style command-line tool for Azure

Uses

- Resource management
- Automation
- Scripting
- Infrastructure deployment

Example

```
az group create
```

Creates a Resource Group.

Exam Tip

If the question mentions:
"Bash commands"
☑ Azure CLI

Azure PowerShell vs Azure CLI

Azure PowerShell	Azure CLI
Uses PowerShell cmdlets	Uses Bash-style commands
Best for PowerShell users	Best for Linux/Bash users
Cross-platform	Cross-platform
Automation & scripting	Automation & scripting

Exam Favorite
Both perform almost the same tasks.
Difference = **command syntax**.

Copilot in Azure
Copilot in Azure is an AI-powered assistant that helps administrators manage Azure resources using natural language.
Simple Definition
Copilot = AI Assistant for Azure

What Can It Help With?
- Explain Azure services
- Generate commands
- Assist with troubleshooting
- Recommend configurations
- Help manage resources

Exam Tip
Copilot provides **guidance and assistance**, but administrators should still verify recommendations.

Common Exam Questions
Which Azure tool provides a graphical interface?
☑ Azure Portal

Which Azure tool is a browser-based shell?
☑ Azure Cloud Shell

Which Azure tool uses cmdlets?
☑ Azure PowerShell

Which Azure tool uses Bash-style commands?
☑ Azure CLI

Which Azure tool provides AI-assisted management?
☑ Copilot in Azure

AZ-900 Quick Revision

Tool	Purpose
Azure Portal	GUI Management
Azure Cloud Shell	Browser-based Command Line
Azure PowerShell	PowerShell Cmdlets
Azure CLI	Bash Commands
Copilot in Azure	AI Assistant

Memory Trick
GUI
🖥 Azure Portal

Browser Shell
☁ Cloud Shell

PowerShell Commands
⚡ Azure PowerShell

Bash Commands
💻 Azure CLI

AI Assistant
🤖 Copilot in Azure

Revision
Azure Portal → GUI
Cloud Shell → Browser-based shell
Azure PowerShell → Cmdlets
Azure CLI → Bash commands
Copilot → AI assistance
👉 **Portal = Click, CLI/PowerShell = Type, Copilot = Ask.** 🚀

Purpose Of Azure Arc:

What is Azure Arc?
Azure Arc is a service that extends Azure management and governance capabilities to resources running:
- On-premises (your own datacenter) (your own datacenter)
- Other cloud providers (AWS, GCP)

- Azure

Simple Definition

Azure Arc = Manage non-Azure resources as if they were Azure resources.

Why Azure Arc?

Organizations often use:

- Azure
- On-premises datacenters
- AWS
- Google Cloud

Managing all these environments separately becomes difficult.

Azure Arc provides a **single management plane** through Azure Resource Manager (ARM).

What Azure Arc Does

Azure Arc lets you:

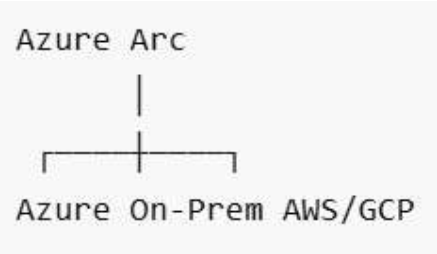
- ☑ Manage hybrid environments
- ☑ Manage multicloud environments
- ☑ Apply Azure Policy everywhere
- ☑ Use Azure monitoring tools everywhere
- ☑ Maintain governance and compliance
- ☑ Manage resources from one portal

Key Idea

Without Azure Arc:

```
Azure → Managed separately
On-Prem → Managed separately
AWS → Managed separately
GCP → Managed separately
```

With Azure Arc:



Everything can be managed from Azure.

How Azure Arc Works

Azure Arc projects non-Azure resources into **Azure Resource Manager (ARM)**.

Once connected, those resources appear in Azure Portal just like native Azure resources.

Then You Can Apply:

- Azure Policy
- Azure Monitor
- RBAC
- Tags
- Defender for Cloud
- Governance controls

Practical Scenario

Imagine a company has:

- Some servers in Azure
- Some servers in its own datacenter
- Some workloads in AWS

Without Azure Arc:

- Three different management systems

With Azure Arc:

- Single Azure management experience

Resources Azure Arc Can Manage

1. Servers

Windows and Linux servers running outside Azure.

2. Kubernetes Clusters

Manage Kubernetes clusters anywhere.

Examples:

- On-prem Kubernetes
- AWS EKS
- Google GKE

3. Azure Data Services

Run Azure data services outside Azure.

Examples:

- Azure SQL Managed Instance
- Azure PostgreSQL

4. SQL Servers

Manage SQL Servers running:

- On-premises
- AWS
- GCP

5. Virtual Machines

Manage VMs running outside Azure.

Benefits of Azure Arc

Unified Management

Single control plane for all environments.

Governance

Apply Azure Policies across clouds and datacenters.

Monitoring

Monitor resources using Azure tools.

Security
Use Microsoft Defender for Cloud for non-Azure resources.

Consistency
Same management experience everywhere.

Exam-Focused Points ⭐
Azure Arc allows Azure management outside Azure.

Azure Arc supports:

- On-premises servers
- Kubernetes clusters
- SQL Servers
- Azure data services
- Virtual machines

Azure Arc uses:
☑ Azure Resource Manager (ARM)

Azure Arc helps with:
☑ Hybrid cloud
☑ Multicloud
☑ Governance
☑ Monitoring
☑ Compliance

Common Exam Questions
What is Azure Arc?
A service that extends Azure management and governance to resources running outside Azure.

Can Azure Arc manage AWS and GCP resources?
☑ Yes

Which Azure component does Azure Arc integrate with?
☑ Azure Resource Manager (ARM)

What problem does Azure Arc solve?
☑ Unified management of hybrid and multicloud environments.

Quick Revision

Feature	Azure Arc
Purpose	Manage resources outside Azure
Supports On-Prem	✅
Supports AWS/GCP	✅
Uses ARM	✅
Supports Azure Policy	✅
Supports Azure Monitor	✅
Supports Defender for Cloud	✅
Hybrid Cloud Solution	✅

Memory Trick
Arc = Azure Reaches Clouds
Think:
👉 Azure Arc extends Azure's management "arc" beyond Azure itself.
Azure Arc = One Dashboard for Azure + On-Prem + Other Clouds ☁️🌍🚀

Azure Resource Manager and Azure ARM Templates:

What is Azure Resource Manager (ARM)?
Azure Resource Manager (ARM) is the **deployment and management service for Azure**.
It provides a management layer that allows you to:

- Create resources
- Update resources
- Delete resources
- Organize resources
- Apply security and governance controls

Simple Definition
ARM = The control plane of Azure.
Whenever you use:

- Azure Portal
- Azure CLI
- Azure PowerShell
- REST APIs
- SDKs

the request first goes through **Azure Resource Manager**.

ARM Request Flow

```
Azure Portal
Azure CLI
PowerShell
REST API
      ↓
Azure Resource Manager (ARM)
      ↓
Azure Services
   - Compute
   - Networking
   - Storage
   - Databases
```

ARM authenticates, authorizes, and processes all requests.

Azure Resource Manager Benefits

1. Consistent Management Layer

All Azure tools communicate through ARM.

Result:

☑ Same functionality across Portal, CLI, PowerShell, APIs

2. Resource Group Management

Manage resources as a group instead of individually.

Example:

```
Resource Group
 ├─ VM
 ├─ Storage Account
 └─ Virtual Network
```

3. Dependency Management

ARM automatically deploys resources in the correct order.

Example:

```
Virtual Network
      ↓
Virtual Machine
```

VM deploys only after VNet is ready.

4. Role-Based Access Control (RBAC)

ARM integrates with Azure RBAC.

You can control who can:

- Read
- Create
- Modify
- Delete

resources.

5. Tag Support

Apply tags for:

- Cost tracking
- Ownership
- Environment

Example:

```
Environment = Production
Owner = IT Team
```

Infrastructure as Code (IaC)

Infrastructure as Code means managing infrastructure using code rather than manually creating resources.

Instead of:

```
Portal → Click → Create VM
```

you define infrastructure in a file:

```
VM
Storage
Network
```

and deploy automatically.

Azure Resource Manager Templates

ARM Templates are **JSON files** that define Azure resources and configurations.

Example

```
</> JSON

{
  "type": "Microsoft.Compute/virtualMachines",
  "name": "MyVM"
}
```

The template describes:

- What resources to create
- Configuration settings
- Dependencies

ARM then deploys everything automatically.

Benefits of ARM Templates

Declarative Syntax

You specify:

☑ What should exist

Not:

❌ Step-by-step deployment instructions

Repeatable Deployments

Use the same template multiple times.
Example:
```
Dev Environment
Test Environment
Production Environment
```
All stay consistent.

Automatic Orchestration
ARM handles:
- Dependencies
- Deployment order
- Parallel deployments

Modularity
Large deployments can be split into smaller templates.

Extensibility
Can integrate with:
- PowerShell scripts
- Bash scripts

for additional setup tasks.

What is Bicep?
Bicep is Microsoft's modern language for deploying Azure resources.
It is built on top of ARM.
Simple Definition
Bicep = Easier and cleaner way to write ARM templates.

ARM Template vs Bicep

ARM Template	Bicep
JSON format	Simple language
Verbose	Concise
Harder to read	Easier to read
More complex	Simpler
Supported by ARM	Supported by ARM

Example
ARM JSON:

```json
{
  "type": "Microsoft.Storage/storageAccounts"
}
```

Bicep:

```
resource storage 'Microsoft.Storage/storageAccounts'
```

Much cleaner.

Benefits of Bicep
Simpler Syntax
Less code to write.

Current Azure Resource Support
Supports latest Azure resource types and API versions.

Repeatable Deployments
Consistent deployments across environments.

Built-in Orchestration
ARM automatically handles dependencies.

Modularity
Reuse deployment logic using modules.

Exam-Focused Points ★
Azure Resource Manager (ARM)
- Deployment and management service for Azure
- All Azure tools communicate through ARM
- Supports RBAC
- Supports Tags
- Supports Resource Groups
- Handles authentication and authorization

ARM Templates
- JSON-based
- Infrastructure as Code (IaC)
- Declarative
- Repeatable deployments
- Dependency management

Bicep
- Declarative language
- Simpler than JSON ARM templates
- Deploys through ARM
- Supports modules and reusable code

Common Exam Questions
What is Azure Resource Manager?

☑ Azure's deployment and management service.

What format do ARM templates use?
☑ JSON

What type of model do ARM templates use?
☑ Declarative model

What is Bicep?
☑ A simpler language used to deploy Azure resources through ARM.

Does Bicep replace ARM?
❌ No
☑ Bicep works on top of ARM and compiles into ARM templates.

Quick Revision

Service	Purpose
Azure Resource Manager (ARM)	Deploy and manage Azure resour
ARM Templates	JSON-based Infrastructure as Cod
Bicep	Simpler ARM template language
Resource Groups	Organize resources
RBAC	Access control
Tags	Resource organization and cost tracking

Memory Trick
ARM = Azure Resource Manager = Azure's Manager
Think:

```
Portal / CLI / PowerShell
          ↓
         ARM
          ↓
   Azure Resources
```

And remember:
ARM Template = JSON IaC
Bicep = Easier ARM Template

Azure Service Health:

What is Azure Service Health?
Azure Service Health helps you monitor the health of Azure services and your Azure resources.
It provides personalized alerts and guidance when Azure issues affect your environment.

Three Health Views
Azure Service Health consists of **three levels of visibility**:

```
Azure Status
     ↓
Service Health
     ↓
Resource Health
```

1. Azure Status
A **global view** of Azure services across all regions.
Purpose
Check whether Azure is experiencing a widespread outage.
Example
You hear that Azure East US is having issues.
Check Azure Status to see:

- Which services are affected
- Which regions are impacted
- Current outage information

Scope

```
All Services
+
All Regions
```

Exam Point ★
Azure Status = Public global view of Azure health

2. Service Health
A personalized view of Azure services you actually use.
Purpose
Shows:

- Service issues
- Planned maintenance
- Health advisories

that affect your subscriptions.
Features
☑ Personalized alerts
☑ Email notifications
☑ Impact analysis
☑ Maintenance notifications
Example
If your organization uses:

- Azure Virtual Machines

- Azure SQL Database

Service Health notifies you only about events affecting those services.
Scope
`Your Services`
+
`Your Regions`
Exam Point ★
Service Health = Personalized health alerts for your Azure environment

3. Resource Health
Health information for an individual Azure resource.
Purpose
Shows whether a specific resource is:

- Available
- Degraded
- Unavailable
- Unknown

Example
A VM stops responding.
Resource Health helps determine:

- Is Azure causing the issue?
- Is it a user configuration issue?
- Is the resource healthy?

Scope
`Single Resource`
Examples:

- Virtual Machine
- Storage Account
- Database

Exam Point ★
Resource Health = Health status of a specific resource

Operational Value
Azure Service Health helps you:

- Detect service disruptions quickly
- Receive alerts automatically
- Track planned maintenance
- Identify Azure-caused problems
- Reduce troubleshooting time

Comparison Table

Service	Scope	Purpose
Azure Status	All Azure services and regions	Global Azure health
Service Health	Your subscriptions and services	Personalized alerts
Resource Health	Individual resource	Diagnose resource issues

Example Scenario
Suppose your web application is down.
Step 1
Check **Resource Health**
`Is my VM healthy?`

Step 2
Check **Service Health**
`Is there a service issue affecting my region?`

Step 3
Check **Azure Status**
`Is there a global Azure outage?`

Exam-Focused Questions ★
Which service provides a global view of Azure outages?
☑ Azure Status

Which service provides personalized alerts for your Azure resources?
☑ Service Health

Which service helps determine whether a VM is healthy?
☑ Resource Health

Which service shows planned maintenance events?
☑ Service Health

AZ-900 Quick Revision

Tool	Remember As
Azure Status	Global Azure view
Service Health	Your services & subscriptions
Resource Health	Individual resource health

Memory Trick

```
Status   = Whole Azure
Health   = Your Azure
Resource = One Resource
```

Important from Exam Perspective 🚨
"Which Azure service tells whether a specific VM is healthy?"
☑ **Resource Health**
"Which service sends alerts when Azure maintenance affects your subscription?"
☑ **Service Health**
"Which service shows global Azure outages?"
☑ **Azure Status**

Azure Monitor:
Azure Monitor is Microsoft's monitoring platform that helps collect, analyze, visualize, and respond to data from:

- Azure resources
- Applications
- Virtual Machines
- Networks
- On-premises resources
- Multi-cloud environments

Simple Definition
Azure Monitor = Monitor everything running in Azure.

What Does Azure Monitor Collect?
Azure Monitor collects three types of data:
Metrics
Numerical values collected over time.
Examples:

- CPU usage
- Memory usage
- Disk IOPS
- Network traffic

Example:
`CPU Usage = 85%`

Logs
Detailed event records.
Examples:

- Login events
- VM restart events
- Application errors

Example:
`User login failed`

Traces
Application request tracking.
Used mainly by:

- Web applications
- APIs

Example:
```
User Request
→ API
→ Database
→ Response
```

Azure Monitor Workflow
```
Resources
     ↓
Azure Monitor
     ↓
Metrics + Logs + Traces
     ↓
Analyze
     ↓
Alerts / Dashboards / Reports
```

Azure Log Analytics
Log Analytics is the tool used to query and analyze Azure Monitor log data.
Purpose
Allows you to:

- Search logs
- Investigate issues
- Analyze trends
- Troubleshoot problems

Example
Find:
`All VM errors in last 24 hours`
or
`Failed login attempts`

Exam Point ⭐
Log Analytics = Query and analyze logs collected by Azure Monitor

Azure Monitor Alerts
What are Alerts?
Alerts notify you when a defined condition occurs.
Example
Create alert:
`CPU > 80%`
When CPU exceeds 80%:
☑ Email sent
☑ SMS sent
☑ Action triggered

Alert Components
Alert Rule
Defines:
`WHEN to trigger`
Example:
`CPU > 80%`

Action Group
Defines:
`WHAT happens next`
Example:
`Send Email`
`Notify Admin`
`Run Automation`

Types of Alerts
Metric Alerts
Based on metrics.
Example:
`CPU Usage > 80%`
Log Alerts
Based on log queries.
Example:
`Multiple failed login attempts`

Exam Point ★
Alert Rule = Condition
Action Group = Response

Application Insights
What is Application Insights?
Application Insights is a feature of Azure Monitor used to monitor applications.
Purpose
Tracks:

- Performance
- Availability
- Failures
- User activity

for web apps and APIs.

What Can Application Insights Monitor?
Request Rates
`How many requests?`

Response Times
`How fast is the application?`

Failure Rates
`How many requests fail?`

Dependency Performance
Tracks:

- Databases
- APIs
- External services

User Activity
Tracks:

- Sessions
- User counts
- Page views

Server Performance
Tracks:

- CPU
- Memory
- Network

Availability Testing
Application Insights can send synthetic test requests.
Purpose:
Check whether application is available.
Example:
`Ping website every 5 minutes`
If unavailable:
Alert administrators.

Real-Life Example
Suppose your company hosts an e-commerce website.
Azure Monitor
Monitors:

- VM health
- CPU
- Network

Log Analytics
Investigates:
`Why checkout failed?`

Alerts
Notifies when:
`CPU > 90%`

Application Insights
Tracks:

```
Page Load Time
API Failures
User Sessions
```

Exam-Focused Points ⭐

Azure Monitor

- Central monitoring solution
- Collects Metrics, Logs, Traces
- Works across Azure, on-premises, and multicloud

Log Analytics

- Analyze log data
- Query Azure Monitor logs

Alerts

- Notify when conditions are met
- Alert Rule + Action Group

Application Insights

- Application performance monitoring (APM)
- Tracks requests, failures, dependencies, and user activity

Common Exam Questions

Which Azure service collects metrics and logs?

☑ Azure Monitor

Which Azure Monitor feature is used to query logs?

☑ Log Analytics

Which Azure Monitor feature sends notifications?

☑ Alerts

Which service monitors application performance?

☑ Application Insights

What are the two main parts of an alert?

☑ Alert Rule and Action Group

Revision Table

Service	Purpose
Azure Monitor	Monitoring platform
Log Analytics	Analyze logs
Alerts	Notifications and actions
Application Insights	Application monitoring

High-Priority Exam Topics (Must Memorize) 🚨

Identity & Security

- Microsoft Entra ID
- MFA
- SSO
- Conditional Access
- RBAC
- Zero Trust
- Defense in Depth

Storage

- Storage Accounts
- Blob Storage
- Storage Tiers (Hot, Cool, Cold, Archive)
- Storage Redundancy (LRS, ZRS, GRS, GZRS)

Networking

- VNet
- VPN Gateway
- ExpressRoute
- DNS

Governance

- Azure Policy
- Resource Locks
- Tags
- Azure Arc
- ARM
- Bicep

Monitoring

- Azure Monitor
- Log Analytics
- Alerts
- Application Insights
- Service Health

Cost Management

- Pricing Calculator
- Cost Management
- Budgets
- Reservations

- Savings Plan
- Spot VM

Requirement	Answer
Copies within **one datacenter**	**LRS**
Protection from **Availability Zone failure**	**ZRS**
Copy in a **secondary region**	**GRS**
Zone protection + secondary region	**GZRS**

Problem	Answer
Datacenter/zone failure	**Availability Zones**
Entire region failure	**Multi-region / geo-redundancy**
Azure service incident	**Service Health**
Resource/application metrics	**Azure Monitor**
Application telemetry	**Application Insights**
Improvement recommendations	**Azure Advisor**
Who can do what	**Azure RBAC**
What configuration is allowed	**Azure Policy**
Prevent deletion/modification	**source Lock**